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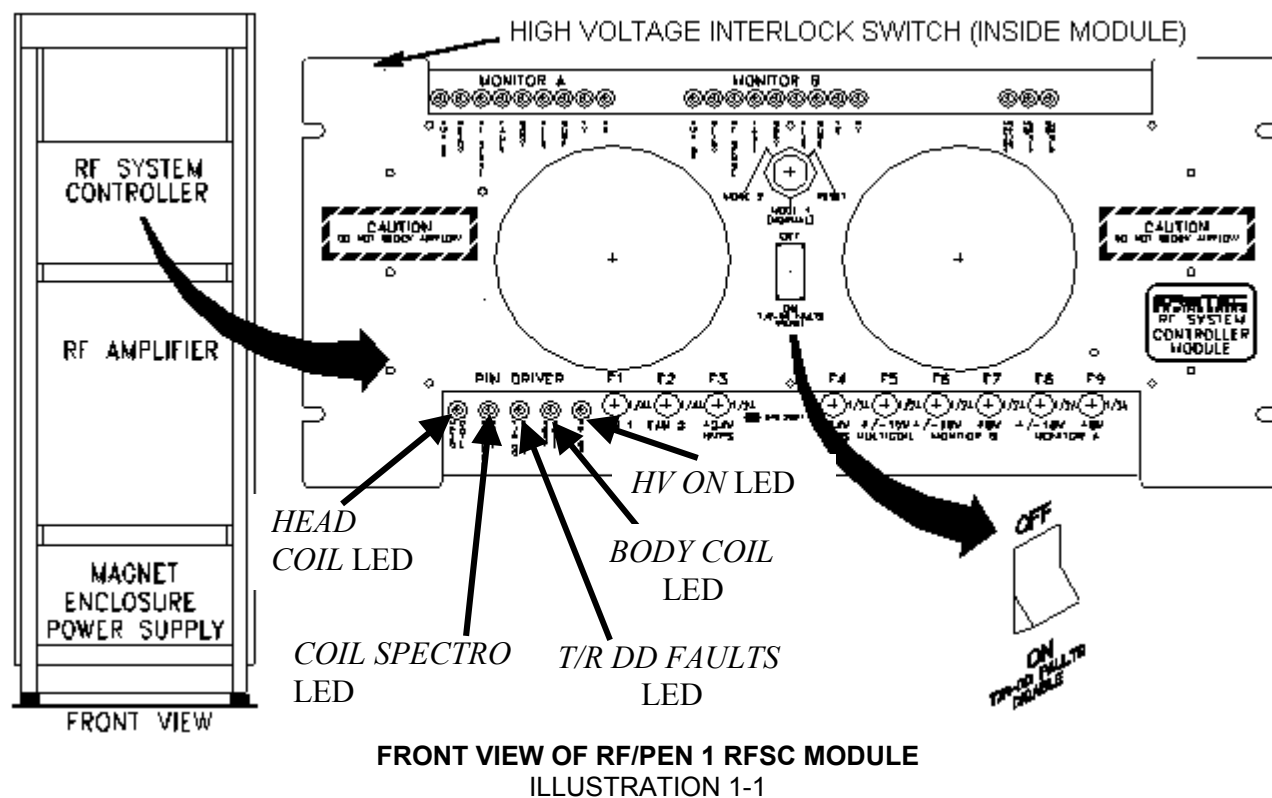
Description - The 1.5T PIN Switch Driver Board of the RF System Controller (RFSC) module provides the dc bias signals needed for switching the various transmit/receive (TR) switches, dynamic disable switches, and direct drive switches. This unit is included in all RF/PEN I cabinets. See Illustration 1-1.

Note

Calibration with Spectro Option - For a Signa 1.5T system with the spectroscopy option, refer to Direction 2151387, Signa Horizon Spectroscopy Subsystem, Set Up and Calibration tab, Section 1, Broadband and Narrow Band RF Cabinets for Broadband RF Amplifier output calibration.

WARNING!

POSSIBLE ELECTRIC SHOCK! USE EXTREME CARE WHEN REMOVING OR REPLACING JUMPERS IN THE MODULE. THERE IS 120 VAC PRESENT INSIDE THE MODULE. WITH THE HIGH VOLTAGE INTERLOCK SWITCH DISABLED AND PULLED INTO THE FULL UP POSITION, THERE IS 1000 VDC AND 120 VAC PRESENT INSIDE THE MODULE. WHEN WORKING ON THE RFSC, CHECK THE "HV ON" LED ON THE FRONT OF THE RFSC TO DETERMINE IF THE 1000VDC IS ENABLED. SEE ILLUSTRATION 1-1.



1 – RFSC PREPARATION

1-1 Tools and Instruments Required

Refer to Table 1-1 for equipment required to complete this calibration.

TABLE 1-1
 REQUIRED TOOLS AND EQUIPMENT

Item	Description	Part Number
1.	100 MHz Scope	46-183029P61
2.	Digital Multimeter (DMM)	46-194427P49
3.	Pot Tweaker	46-194427P361

1-2 RFSC Removal

1. Remove the 4 screws that secure the front of the RF System Controller (RFSC) unit to the cabinet. See Illustration 1-2.

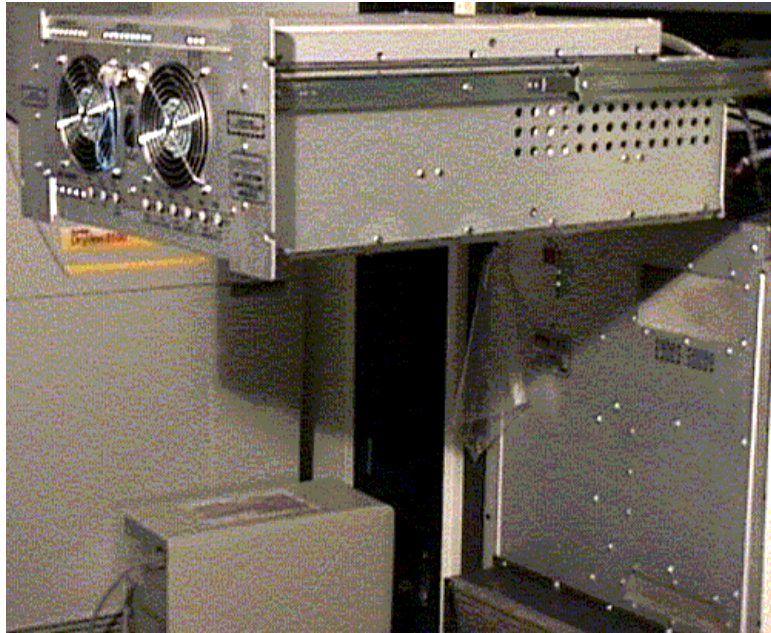


Remove 4 screws

REMOVAL OF SCREWS FROM FRONT OF RFSC
 ILLUSTRATION 1-2

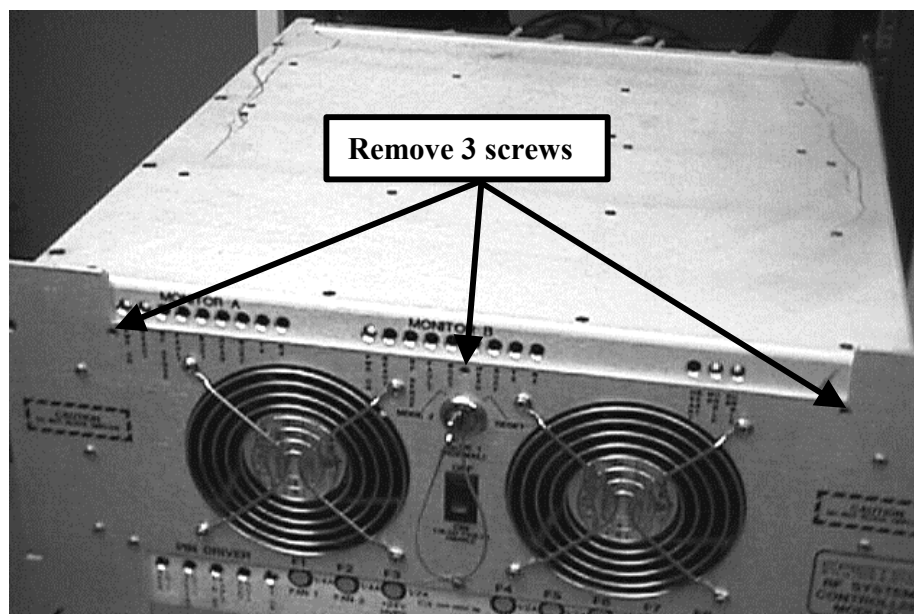
1-2 RFSC Removal (Continued)

2. The RFSC is mounted on slides. Carefully slide the RFSC out of the front of the rack, while minding the cables attached to the rear of the unit, as far as it will go until it locks into place. See Illustration 1-3.



RFSC MODULE FULLY EXTENDED FROM THE FRONT OF THE RF/PEN 1 CABINET
ILLUSTRATION 1-3

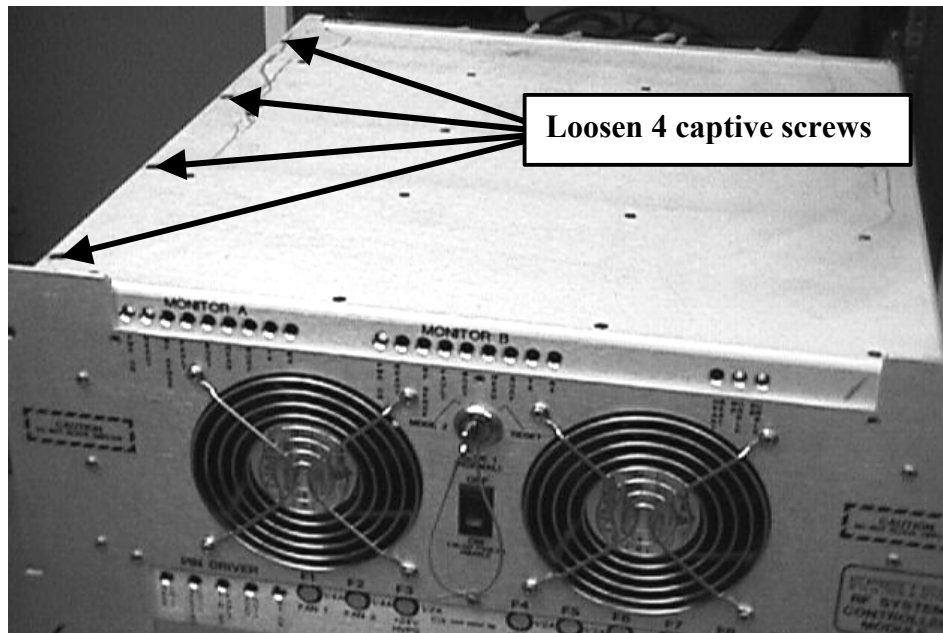
3. Remove the 3 screws above the front fans securing the top cover to the front of the RFSC. See Illustration 1-4.



FRONT RFSC SCREWS
ILLUSTRATION 1-4

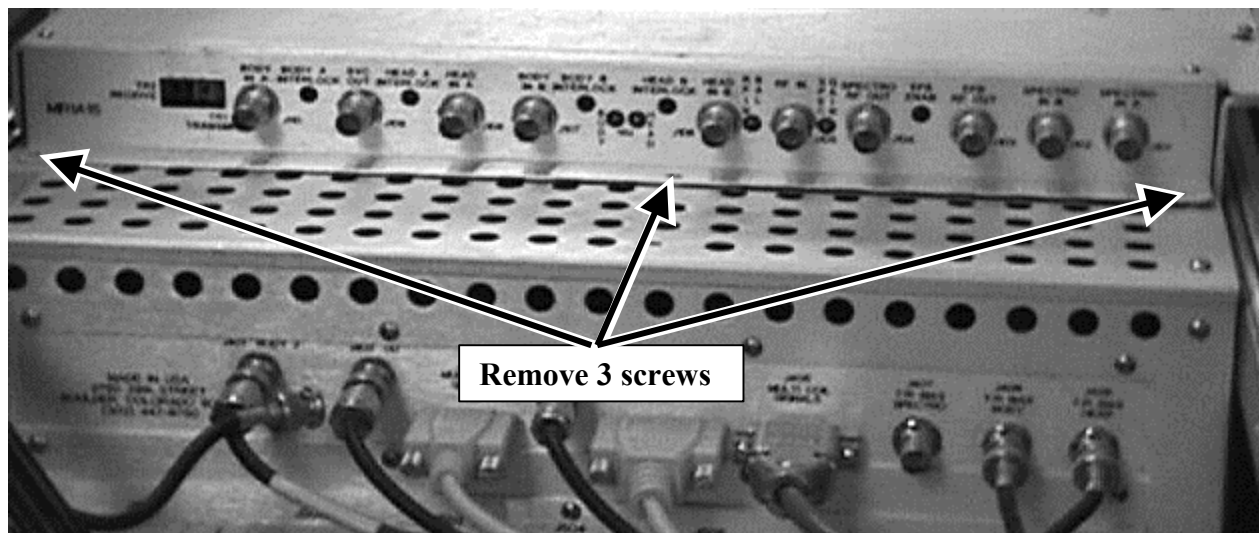
1-2 RFSC Removal (Continued)

4. Loosen the 4 captive screws securing the left side of the top cover. See Illustration 1-5.



CAPTURED SCREWS ON TOP OF RFSC
 ILLUSTRATION 1-5

5. At the rear of the RFSC, remove the 3 screws that secure the top cover to the perforated rear shield plate. See Illustration 1-6.

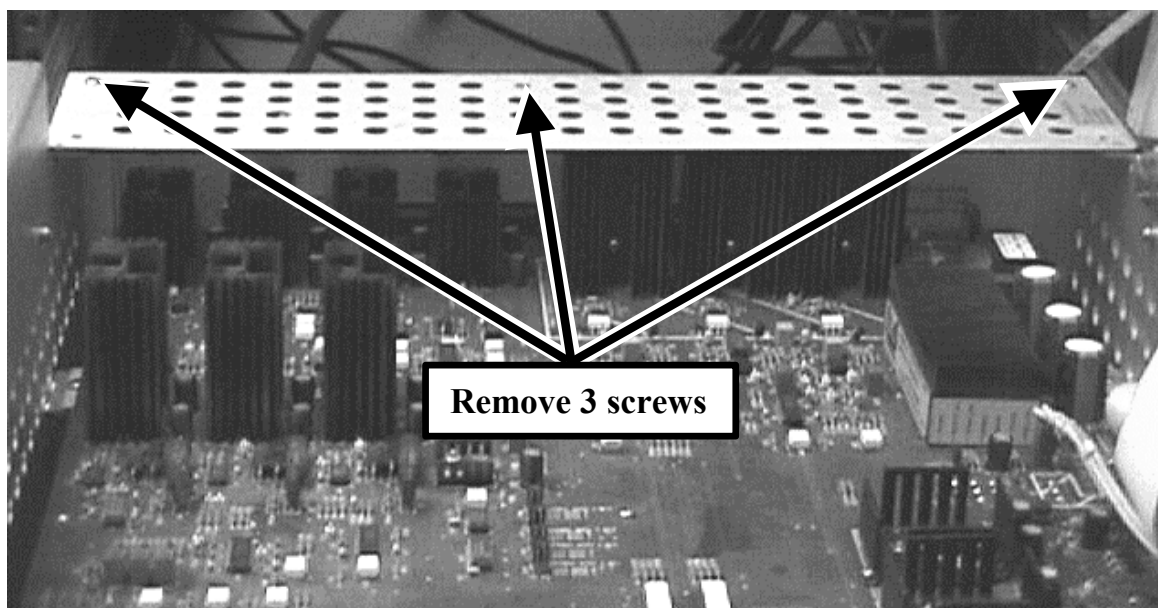


REAR SHIELD PLATE AND TOP COVER (CABLES REMOVED)
 ILLUSTRATION 1-6

6. The top cover is hinged on the right side. Tilt the cover up and beyond 90 degrees into the service position.

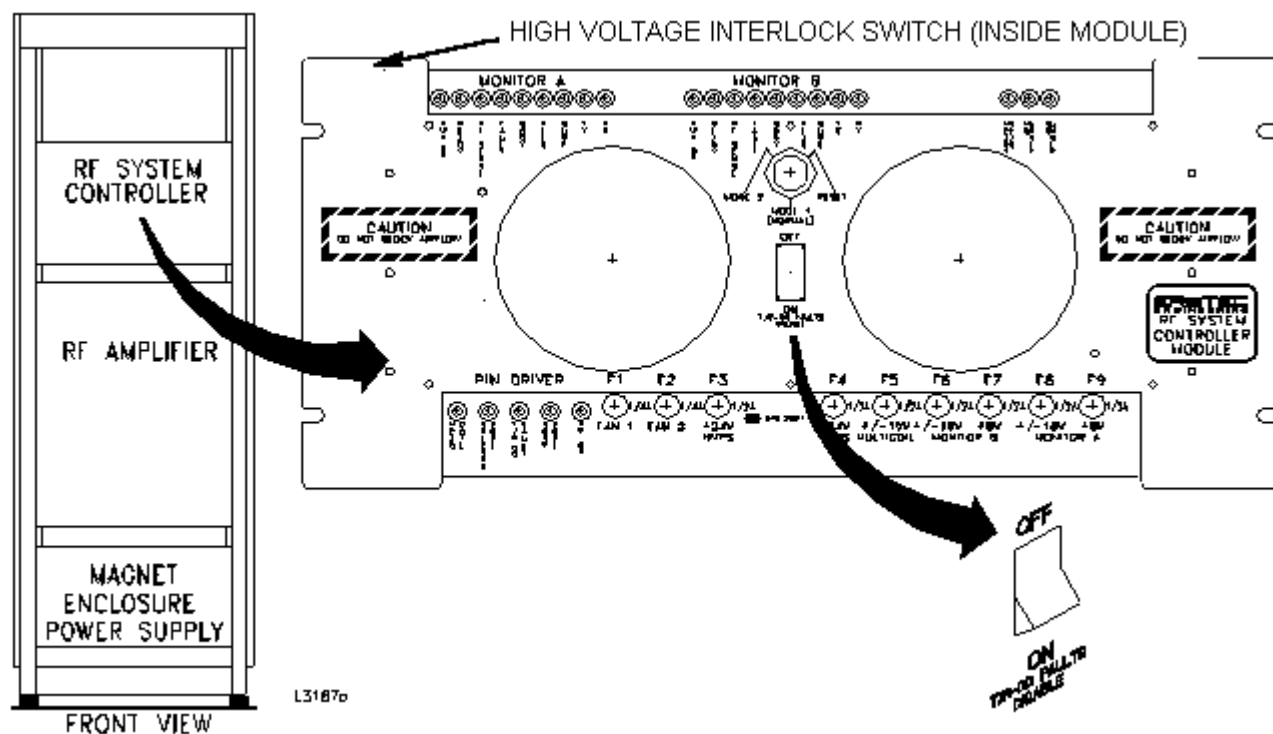
1-2 RFSC Removal (Continued)

7. Remove the 3 screws from the rear shield located at the rear of the RFSC. Remove the shield to obtain access to the test points. See Illustration 1-7.



REMOVING 4 SCREWS FROM REAR SHIELD LOCATED ON TOP REAR OF RFSC
ILLUSTRATION 1-7

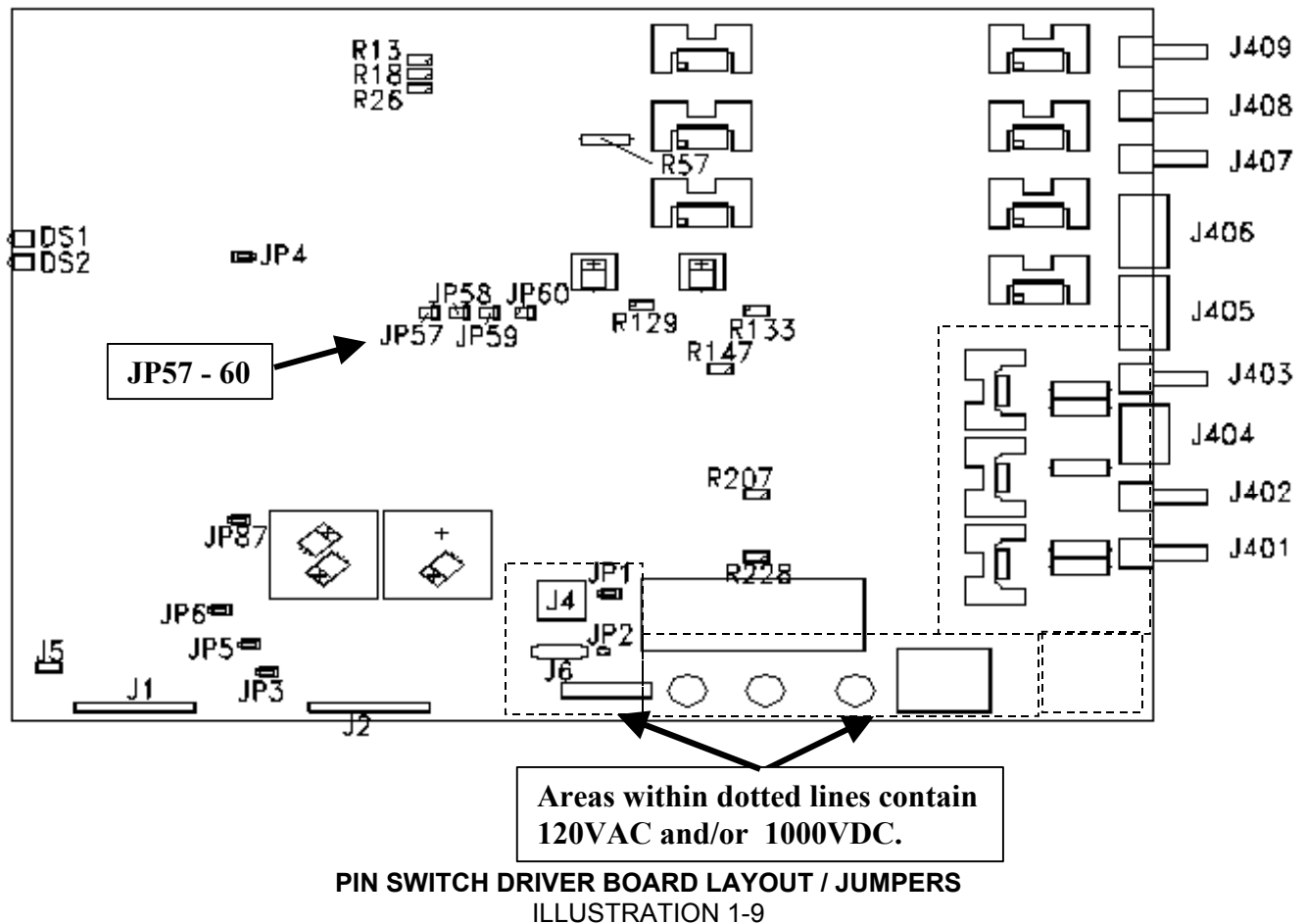
8. Verify that T/R-DD Faults Disable switch is off (faults not disabled) on the RFSC module (MR1A15). See Illustration 1-8.



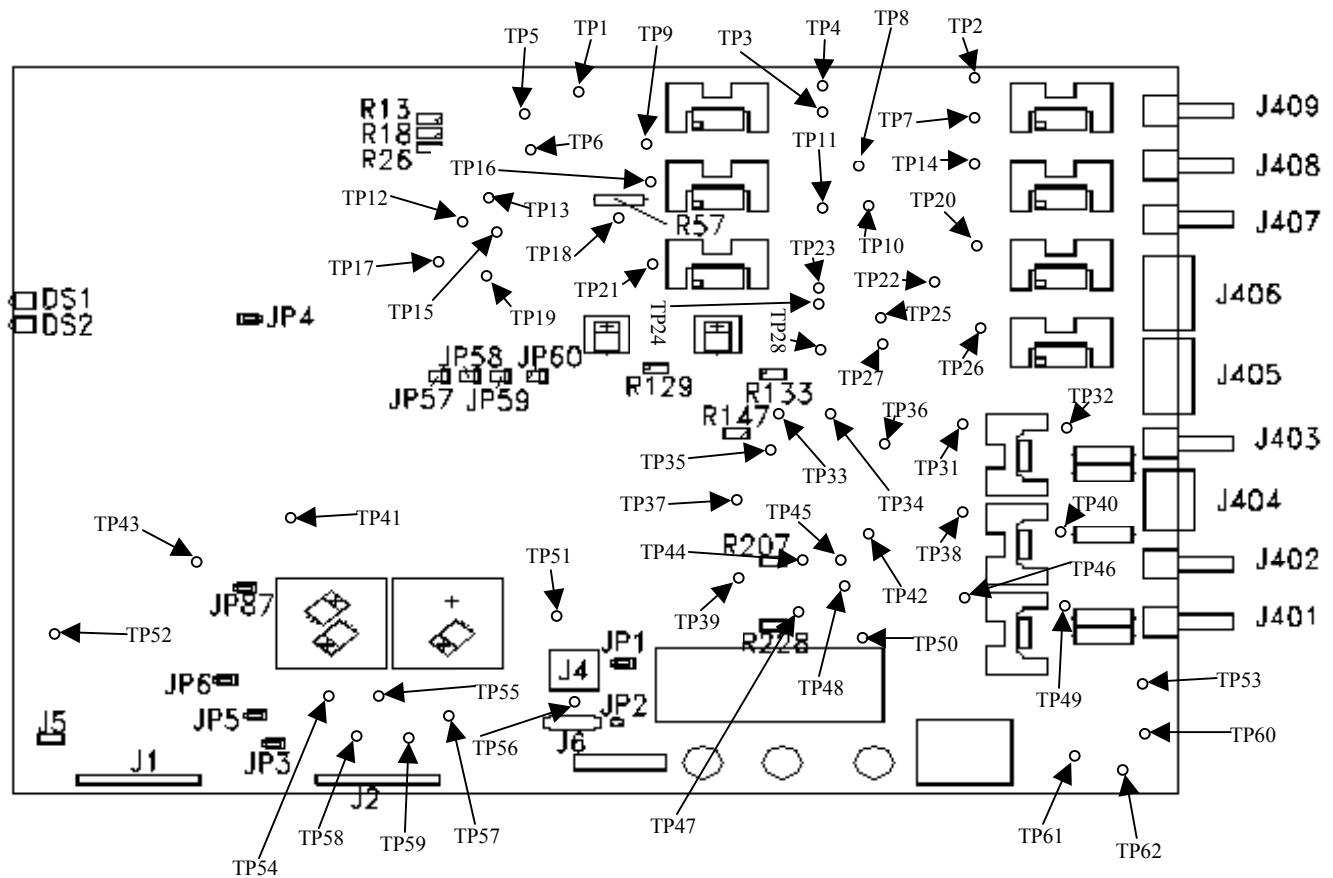
CONFIRM T/R-DD FAULTS DISABLE SWITCH IS IN OFF (UP) POSITION
ILLUSTRATION 1-8

1-3 PIN Switch Driver Board Setup

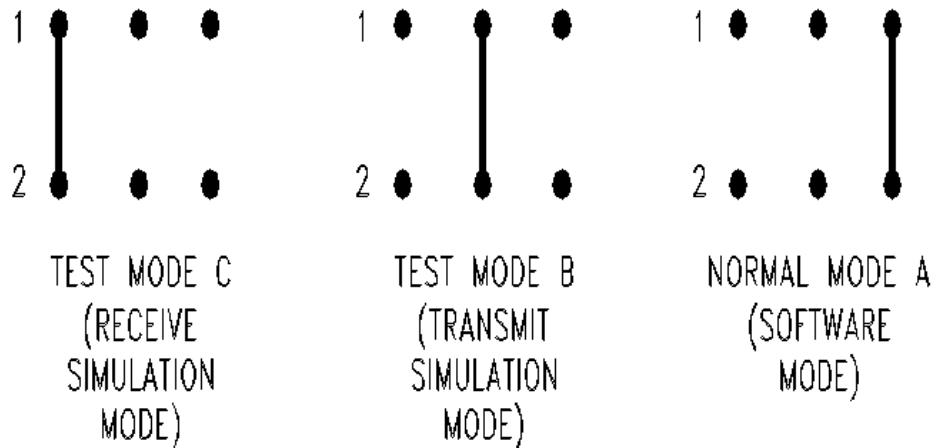
1. Verify that jumpers JP57–60 are in software control mode (normal mode, pins 5 and 6 connected) for four-channel multicoil systems, or in b mode (receive mode, pins 3 and 4 connected) for non-multicoil systems. For two-channel multicoil systems, verify that JP57 and JP59 are in software control mode (normal mode, pins 5 and 6 connected), and that JP58 and JP60 are in b mode (receive mode, pins 3 and 4 connected). See Illustration 1-9 and Illustration 1-11.



1-2 PIN Switch Driver Board Setup (Continued)



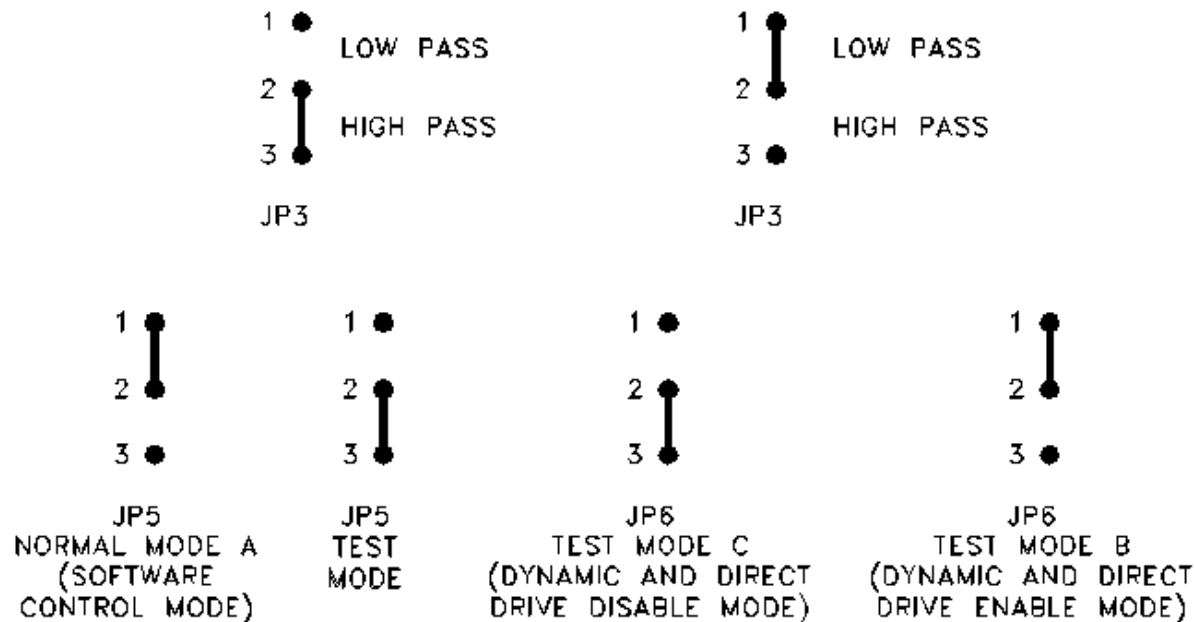
PIN SWITCH DRIVER BOARD TEST POINT LOCATIONS
 ILLUSTRATION 1-10



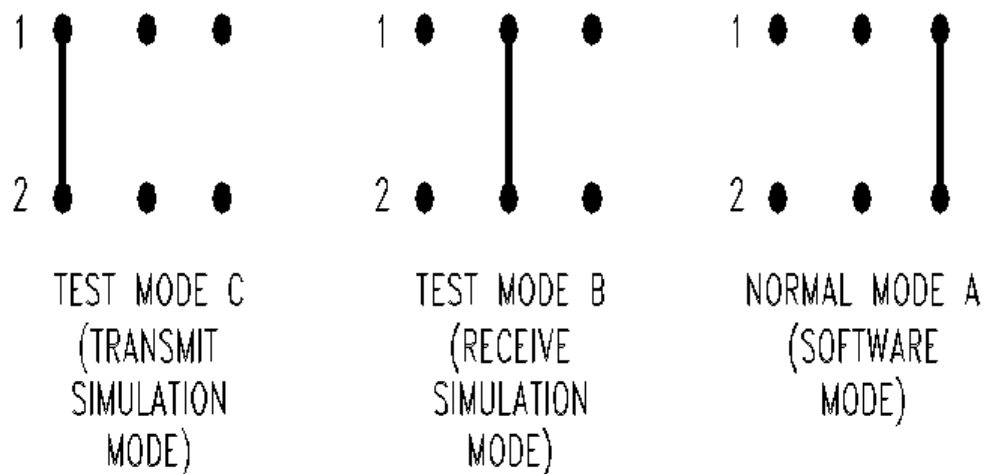
JUMPER JP57-JP60 MODE POSITIONS
 ILLUSTRATION 1-11

1-2 PIN Switch Driver Board Setup (Continued)

- Examine the board and note whether jumper **JP87** is **missing** pins 1 and 2. Verify that JP87 is in software control mode (normal mode, pins 5 and 6 connected). See Illustration 1-13. Verify that JP5 is in normal mode, (pins 1 and 2 connected). See Illustrations 1-9 and 1-12.



JUMPER JP5, JP6, AND JP3 MODE POSITIONS
 ILLUSTRATION 1-12



JUMPER JP87 MODE POSITIONS (PINS 1 AND 2 MAY BE MISSING)
 ILLUSTRATION 1-13

1-2 PIN Switch Driver Board Setup (Continued)

3. Verify that jumper JP3 is in high pass mode. See Illustrations 1-9 and 1-12.
4. Verify that jumper JP1 is in 1000V mode for 1.5T systems, or in 500V mode for 1.0T systems. See Illustration 1-9.

2 – DIRECT DRIVE & BODY COIL DISABLE FUNCTIONAL CHECKS AND ADJUSTMENTS

This section consists of **Section 2-1 Functional Checks**, which should be done at installation or when troubleshooting, and **Section 2-2 Open Circuit Threshold Adjustments**, which should be done at installation and after hardware has been replaced. All these sections will require the FE to measure test points on the PIN Switch Driver Board. Reference Illustration 1-10 to locate the PIN Switch Driver Board test points. If needed, Table A-1 in Appendix A provides a summary of all the test points and corresponding adjustment potentiometers and test point specifications.

2-1 Functional Checks



Measurements should be made with reference to the correct ground test point, and not to the module chassis, unless otherwise stated. Incorrect measurement readings could occur if made with reference to the wrong ground test point.

1. Verify that T/R-DD faults disable switch is off (faults not disabled) on the RFSC module. See Illustration 1-8.
2. Move JP5 and JP6 across pins 2 and 3 (test mode and test mode "C"). See Illustration 1-12. Ensure that jumper JP3 is across pins 2 and 3 (high pass mode).

Note

Body Coil LED - The Body Coil LED at the front of the RFSC module should be off. The Head Coil LED will be on. This sets a negative output voltage for the direct drive disable circuit that simulates the operation of this circuit during head mode. The T/R DD Faults LED will be on because the 1000VDC high voltage is disabled due to the fact that the High Voltage Interlock switch is engaged.

3. Check the Direct Drive and Body Coil Output Current on the PIN Switch Driver Board as follows: Refer to Table 2-1.

TABLE 2-1
 DIRECT DRIVE AND DYNAMIC DISABLE DRIVER (LOADED) CALIBRATION DATA

Step	Function	Location	Pot. Adj.	Specification
3a	Direct Drive Output (J402 Connected)	TP40 (W/ TP56)	None	+1.0 $\pm$ 0.2 Vdc
4a	Dyn. Dis. Driver #1 (J403 Connected)	TP32 (W/ TP56)	None	+1.0 $\pm$ 0.2 Vdc
5a	Dyn. Dis. Driver #2 (J401 Connected)	TP49 (W/ TP56)	None	+1.0 $\pm$ 0.2 Vdc

- a. Measure the voltage at TP40 with respect to -15V_A (TP56). It should be 1.0 $\pm$ 0.2Vdc. The output current is equal to this voltage.
4. Check Dynamic Disable Driver #1 (Body Coil #1) output current on PIN Switch Driver Board as follows: Refer to Table 2-1 above.
 - a. The voltage at TP32 with respect to -15V_A (TP56) should be 1.0 $\pm$ 0.2 Vdc. The output current is equal to twice this voltage.

2-1 Functional Checks (Continued)

5. Check Dynamic Disable Driver #2 (Body Coil #2) output current on PIN Switch Driver Board as follows: Refer to Table 2-1 above.
 - a. The voltage at TP49 with respect to $-15V_A$ (TP56) should be 1.0 ± 0.2 Vdc. The output current is equal to twice this voltage.
6. Measure the loaded Dynamic Disable Driver and Direct Drive voltages on the PIN Switch Driver Board as follows: Refer to Table 2-2.

TABLE 2-2
DYNAMIC DISABLE DRIVER CALIBRATION DATA

Step	Function	Location (Silk Screen)	Pot. Adj.	Specification
6a	Dyn. Dis. Driver #1(Loaded)	TP31 (W/TGND, TP59)		-4.0 ± 1.5 Vdc
6b	Dyn. Dis. Driver #2 (Loaded)	TP46 (W/TGND, TP59)		-4.0 ± 1.5 Vdc
6c	Direct Drive Output (Loaded)	TP38 (W/TGND, TP59)		-2.5 ± 1.0 Vdc

Note

Dynamic Disable Driver #1 and #2 Output Voltages Vary - The output voltages of the drivers may vary from front to back due to differences in the coil loading.

- a. Measure Driver #1 output voltage at TP31 with respect to TGND (TP59). Voltage should equal -4.0 ± 1.5 Vdc .
 - b. Measure Driver #2 output voltage at TP46 with respect to TGND (TP59). Voltage should equal -4.0 ± 1.5 Vdc .
 - c. Measure the Direct Drive output voltage at TP38 with respect to TGND (TP59). Voltage should equal $-2.5V \pm 1.0$ Vdc .
7. Move jumper JP6 across pins 1 and 2 (Dynamic & Direct Drive Enable, Test Mode "B") on the PIN Switch Driver Board. See Illustration 1-12.

Note

Body Coil LED Check - The Body Coil LED on the front panel of the RFSC module should be lit. The Head Coil LED will also be on. This is normal. This sets a positive output voltage for the dynamic disable drive circuits, which simulates the operation of these circuits during body mode. The T/R DD Faults LED will be on because the 1000VDC high voltage is disabled due to the fact that the High Voltage Interlock switch is engaged.

2-1 Functional Checks (Continued)

WARNING!

POSSIBLE ELECTRIC SHOCK! THE NEXT STEP WILL DISABLE THE HIGH VOLTAGE INTERLOCK SWITCH AND THEREBY PLACE 1000VDC ACROSS THE DYNAMIC DISABLE AND DIRECT DRIVE OUTPUTS, BETWEEN TP38 AND TP62 TO GROUND, AND ON CERTAIN COMPONENTS IN THE HIGH VOLTAGE SECTION ON THE PIN SWITCH DRIVER BOARD. EXERCISE CARE WHEN MEASURING FOR 1000VDC AT TP38 AND TP62. AVOID TOUCHING COMPONENT HEATSINKS AND CASES IN THE HIGH VOLTAGE SECTION OF THE PIN SWITCH DRIVER BOARD WHICH MAY ALSO BE AT 1000VDC POTENTIAL.

8. Refer to Illustration 1-8 and place 1000VDC bias on the PIN Switch Driver Board by pulling the white, HV Interlock switch fully up so that it latches into the disabled, up position. It should protrude above the level of the left side of the case. The *HV ON* LED should illuminate after about 5 seconds.
9. Carefully measure TP62 with reference to GND (TP59) and note the voltage for reference in the next step.
10. Measure the Dynamic Disable Driver and Direct Drive output voltages on PIN Switch Driver Board at the following points: Refer to Table 2-4.

TABLE 2-4
DYNAMIC DISABLE DRIVER OUTPUT VOLTAGE CALIBRATION DATA

Step	Function	Location (Silk Screen)	Pot. Adj.	Specification
10a	Dyn. Dis. Driver #1 Output Voltage	TP31(W/ TP59)		Value @ TP62 +0/–10.0 Vdc
10b	Dyn. Dis. Driver #2 Output Voltage	TP46 (W/ TP59)		Value @ TP62 +0/–10.0 Vdc
10c	Direct Drive Output Voltage	TP38 (W/ TP59)		Value @ TP62 +0/–10.0 Vdc

Note

Voltage Values - Voltage should be no less than the value at TP62 +0/–10 Vdc.

- a. Dynamic Disable Driver # 1 at TP31.
 - b. Dynamic Disable Driver # 2 at TP46.
 - c. Direct Drive at TP38.
11. Enable the High Voltage Interlock Switch and remove the High Voltage (1000VDC) bias from the PIN Switch Driver Board by pushing the white switch down so that it unlatches and drops below the level of the left side of the RFSC case. Verify that after 5 seconds the *HV ON* LED goes off.

12. Move JP6 back to test mode "C" (across pins 2 and 3).

2-2 Open Circuit Threshold Adjustment

1. Verify that T/R-DD faults disable switch is off (faults not disabled) on the RFSC module. See Illustration 1-8.
2. Move JP5 and JP6 across pins 2 and 3 (test mode and test mode "C"). See Illustration 1-12. Ensure that jumper JP3 is across pins 2 and 3 (high pass mode).

Note

Body Coil LED - The Body Coil LED at the front of the RFSC module should be off. The Head Coil LED will be on. This sets a negative output voltage for the direct drive disable circuit that simulates the operation of this circuit during head mode. The T/R DD Faults LED will be on because the 1000VDC high voltage is disabled due to the fact that the High Voltage Interlock switch is engaged.

3. Adjust the Direct Drive Open Circuit Detection Level on the PIN Switch Driver Board as shown in Table 2-3.

TABLE 2-3
OPEN CIRCUIT DETECTION CALIBRATION DATA

Step	Function	Location (Silk Screen)	Pot. Adj.	Specification
3b	DD	TP44 (W/TGND,TP59)	R207	TP44 = (TP45 -2) $\pm$ 0.1 V
4b	Body 1	TP34 (W/TGND,TP59)	R147	TP34 = (TP33 -2) $\pm$ 0.1 V
4f	Body 2	TP48 (W/TGND,TP59)	R228	TP48 = (TP47 -2) $\pm$ 0.1 V

- a. Measure the voltage at TP45 with respect to TGND (TP59).
- b. Measure the voltage at TP44 with respect to TGND (TP59). This is the open circuit detection level for this circuit.
- c. Adjust R207 until voltage at TP44 is -2 ± 0.1 Vdc more negative than the measurement taken at TP45.
4. Adjust the Dynamic Disable Driver Open Circuit Detection Levels on PIN Switch Driver Board as shown in Table 2-3 above.
 - a. Measure the voltage at TP33 with respect to TGND (TP59).
 - b. Measure the voltage at TP34 with respect to TGND (TP59). This is the open circuit detection level for Dynamic Disable Driver #1.
 - c. Adjust R147 until voltage at TP34 is -2 ± 0.1 Vdc more negative than the measurement taken at TP33.
 - d. Measure the voltage at TP47 with respect to TGND (TP59).
 - e. Measure the voltage at TP48 with respect to TGND (TP59). This is the open circuit detection level for Dynamic Disable Driver #2.

- f. Adjust R228 until voltage at TP48 is -2 ± 0.1 Vdc more negative than the measurement taken at TP47.

2-2 Open Circuit Threshold Adjustments (Continued)

5. Move JP5 and JP6 across pins 1 and 2 (Normal mode "A" and Test Mode "B"). See Illustration 1-12.

3 – TR DRIVER CIRCUIT ADJUSTMENTS

This section and its subsections should be done at installation and whenever hardware is replaced. Later PIN Switch Driver boards may be missing pins 1 and 2 of JP87. This was done by the vendor to prevent damage to board components resulting from the board being left in transmit mode for extended periods of time. Boards with pins 1 and 2 of JP87 present can be calibrated using **Section 3-1 TR Driver Circuit Adjustments For Boards With JP87 Pins 1 and 2**. Boards that are missing pins 1 and 2 of JP87 should be calibrated using **Section 3-2 TR Driver Circuit Adjustments For Boards Without JP87 Pins 1 and 2**. Reference Illustration 1-10 to locate the PIN Switch Driver Board test points. If needed, Table A-1 in Appendix A provides a summary of all the test points and corresponding adjustment potentiometers and test point specifications.

3-1 TR Driver Circuit Adjustments For Boards With JP87 Pins 1 and 2

1. Be sure that T/R-DD Faults Disable switch is off (faults not disabled) on front panel of RFSC module. See Illustration 1-8.



For the following measurements of TP16, TP9 and TP21 as shown in Table 3-1 do not leave jumper JP87 across pins 1 and 2 (Transmit Simulation, Test Mode "C") for extended periods of time. Equipment damage may result. Take note that steps 3b, 3d, and 3f will direct the FE to move the jumper across pins 5 and 6 after each measurement is completed.

2. Connect DVM leads to TP16 with respect to TGND (TP59). Move jumper JP87 across pins 1 and 2 (Transmit Simulation, Test Mode "C"). See Illustration 1-13. This sets a positive output voltage for the TR driver circuits which simulates the transmit mode.
3. Adjust the loaded TR driver output voltages on PIN Switch Driver Board as shown in Table 3-1.

TABLE 3-1
TR DRIVER CALIBRATION DATA

Step	Function	Location (Silk Screen)	Pot. Adj.	Specification
3a	TR Driver (Body) (Loaded)	TP16 (W/TP59)	R13	+4.0 $\pm$ 0.1 Vdc
3c	TR Driver (Head) (Loaded)	TP9 (W/TP59)	R26	+8.0 $\pm$ 0.1 Vdc
3e	TR Driver (Spectro) (Loaded)	TP21 (W/TP59)	R18	+4.3 $\pm$ 0.1 Vdc

- a. Measure the body TR driver output voltage at TP16.
- b. Adjust R13 until the voltage at TP16 is +4.0 $\pm$ 0.1 Vdc. Move jumper JP87 across pins 5 and 6 (software mode).

3-1 TR Driver Circuit Adjustments For Boards With JP87 Pins 1 and 2 (Continued)

- c. Connect DVM leads to TP9 with respect to TGND (TP59). Move jumper JP87 across pins 1 and 2 (Transmit Simulation, Test Mode "C"). Measure the Head TR Driver output voltage at TP9.
- d. Adjust R26 until the voltage at TP9 is 8.0 ± 0.1 Vdc. Move jumper JP87 across pins 5 and 6 (software mode).

Note

If the system has spectroscopy hardware then connect the Spectro TR switch box to the connectors on the head carriage before performing the next step. Otherwise, skip the next 2 steps.

- e. Connect DVM leads to TP21 with respect to TGND (TP59). Move jumper JP87 across pins 1 and 2 (Transmit Simulation, Test Mode "C"). Measure the Spectro TR Driver output voltage at TP21.
- f. Adjust R18 until the voltage at TP21 is 4.3 ± 0.1 Vdc. Move jumper JP87 across pins 5 and 6 (software mode).



For the following measurements of TP13, TP1 and TP18 as shown in Table 3-2 do not leave jumper JP87 across pins 1 and 2 (Transmit Simulation, Test Mode "C") for extended periods of time. Equipment damage may result. Take note that each of the steps 5 – 7 will direct the FE to move the jumper across pins 5 and 6 after each measurement is completed.

- 4. Verify that the loaded TR driver circuit operation on PIN Switch Driver Board as shown in Table 3-2.

TABLE 3-2
TR DRIVER UNLOADED/LOADED CALIBRATION DATA

Step	Function	Location (Silk Screen)	Pot. Adj.	Specification
5	TR Driver (Body) (Loaded)	TP13 (W/TP59)		> Value @ TP12
6	TR Driver (Head) (Loaded)	TP1 (W/TP59)		> Value @ TP17
7	TR Driver (Spectro) (Loaded)	TP18 (W/TP59)		> Value @ TP19

- 5. Move jumper JP87 across pins 1 and 2 (Transmit Simulation, Test Mode "C"). Verify that the voltage at TP13 is greater than the voltage at TP12, the body TR threshold. Move jumper JP87 across pins 5 and 6 (software mode). The voltage at TP13 is directly proportional to the body TR switch load current: **Current (I) of body TR = Voltage of TP13 ÷ 10**. If TP13 is not greater than TP12, recheck measurement at TP12.
- 6. Move jumper JP87 across pins 1 and 2 (Transmit Simulation, Test Mode "C"). Verify that the voltage at TP1 is greater than the voltage measured at TP17, the head TR threshold. Move jumper JP87 across pins 5 and 6 (software mode). The voltage at TP1 is directly proportional to the head TR switch load current: **Current (I) of head TR = Voltage of TP1 ÷ 10**. If TP1 is not greater than TP17, recheck measurement TP17.

3-1 TR Driver Circuit Adjustments For Boards With JP87 Pins 1 and 2 (Continued)

Note

If the system has spectroscopy hardware then connect the Spectro TR switch box to the connectors on the head carriage before performing the next step. Otherwise, skip the next step.

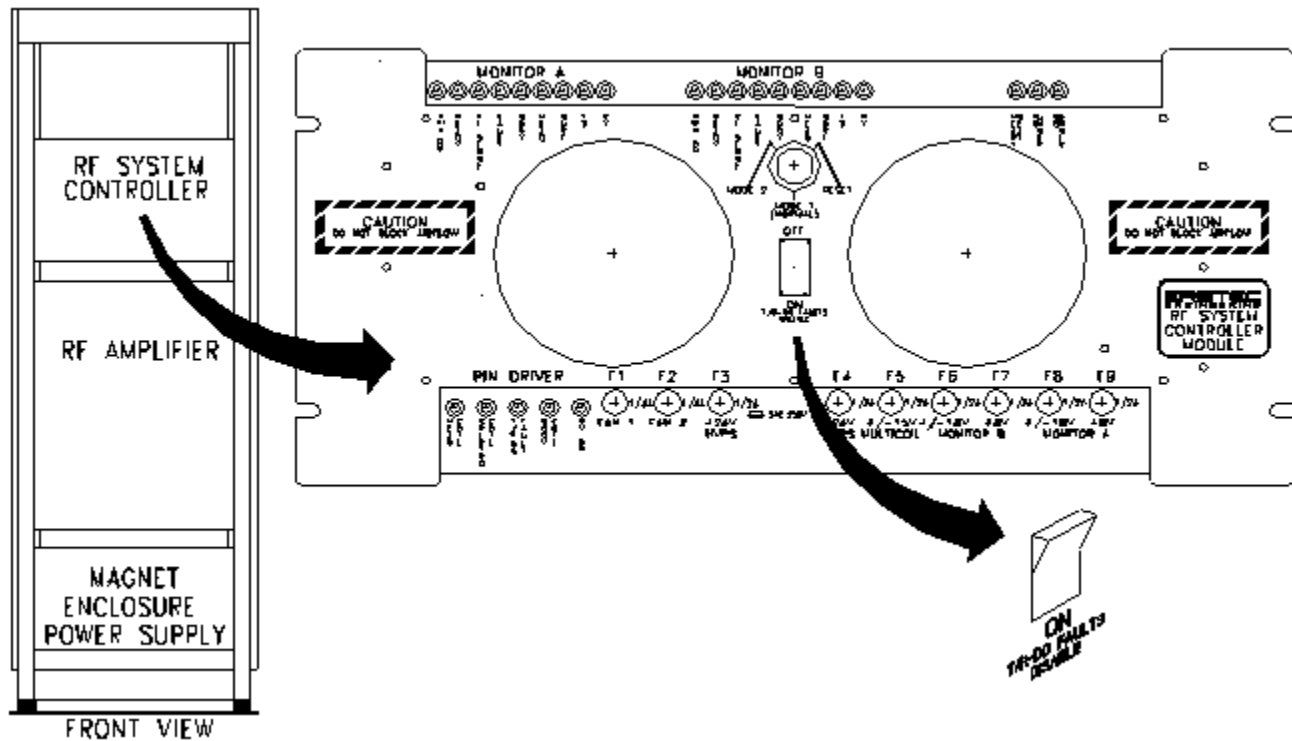
7. Move jumper JP87 across pins 1 and 2 (Transmit Simulation, Test Mode "C"). Verify that the voltage at TP18 is greater than the voltage measured at TP19, the spectroscopy TR threshold. Move jumper JP87 across pins 5 and 6 (software mode). The voltage at TP18 is directly proportional to the Spectro TR switch load current: **Current (I) of spectro TR = Voltage of TP18 ÷ 10**. If Spectro TR Switch Box is not installed, then you will read less than 1.0 Vdc due to lack of current draw. If Spectro TR Switch is installed, and TP18 is not greater than TP19, recheck measurement at TP19.
8. Reset any power monitor faults visible from the fault LEDs on the front of the top cover.
9. Verify that JP5 and JP6 are connected across pins 1 and 2 (Normal mode "A" and Test Mode "B") and that JP87 is connected across pins 5 and 6 (software mode) on the PIN Switch Driver board. See Illustrations 1-12 and 1-13.
10. Verify the dynamic operation of the TR drivers:
 - a. Verify that the switch on the front of the ISE CERD (MR2A30A13) is set to "RF Out Disable" to remove excitation from the RF amplifier. The switch is located just above the J109 connector.
 - b. Move T/R-DD Faults Disable switch on the front of the RFSC to the off position. See Illustration 1-8.
 - c. Set up for a body coil scan.
 - d. Select [**Manual Prescan**], then select [**Scan TR**].
 - e. If TR errors are reported, and TR Switches are verified to be OK, recheck connections to J407, J408, and J409 at the rear of the RFSC module, and verify the sections of the calibration procedure relating to the TR driver circuits.
 - f. When operation is satisfactory, select [**Stop Scan**].
 - g. Set the switch on the front of the ISE CERD (MR2A30A13) to "RF Out Normal" to enable excitation to the RF amplifier.

3-2 TR Driver Circuit Adjustments For Boards Without JP87 Pins 1 and 2

1. At ISE CERD (MR2A30A13) move RF toggle switch, located just above the J109 connector, to "RF Out Disable" to remove excitation from RF amplifier.
2. Verify that jumper JP87 is across pins 5 and 6 in Normal Mode A (software mode). See Illustration 1-13.

3-2 TR Driver Circuit Adjustments For Boards Without JP87 Pins 1 and 2 (Continued)

3. Move T/R-DD Faults Disable switch to the ON position (service mode) on the RFSC module face panel to disable error reporting from the board. See Illustration 3-1.

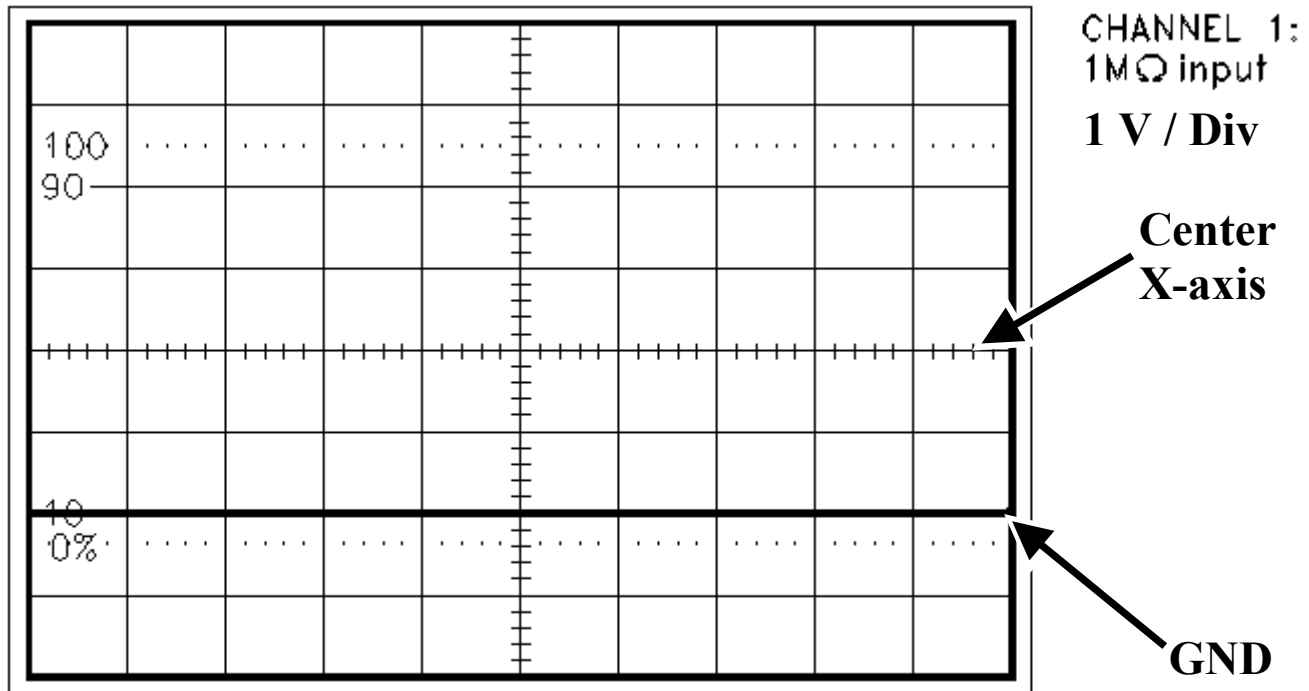


DYNAMIC DISABLER / TR DRIVER IN TEST POSITION
ILLUSTRATION 3-1

4. Setup an oscilloscope near the RFSC and set the input voltage scale on channel 1 to 1Volt / Division.

3-2 TR Driver Circuit Adjustments For Boards Without JP87 Pins 1 and 2 (Continued)

5. Set the channel 1 input selector switch to ground and adjust the scope vertical position control so that the horizontal line trace, instead of being centered on the center horizontal X-axis line on the scope face, is now exactly 2 divisions below it. See Illustration 3-2.



RE-POSITIONING CHANNEL 1 ZERO REFERENCE
ILLUSTRATION 3-2

6. Set the channel 1 input selector switch from ground to DC.
7. Adjust the TR drivers dynamically as follows:
 - a. Set up and landmark for a body coil scan. Any body protocol will work but service protocol o.39 meant for FID Tuning could be used.
 - b. Select **[Manual Prescan]**, then select **[Scan TR]**.
 - c. Attach the scope leads to TP16 with the reference to the RFSC chassis and trigger the signal so that there is one pulse exhibited on the scope.

3-2 TR Driver Circuit Adjustments For Boards Without JP87 Pins 1 and 2 (Continued)

- d. Refer to Table 3-3 and adjust R13 until the voltage 2 msec after the leading edge of the waveform at TP16 is $+4.0 \pm 0.1$ VDC with respect to the ground reference on the scope. See Illustration 3-3.

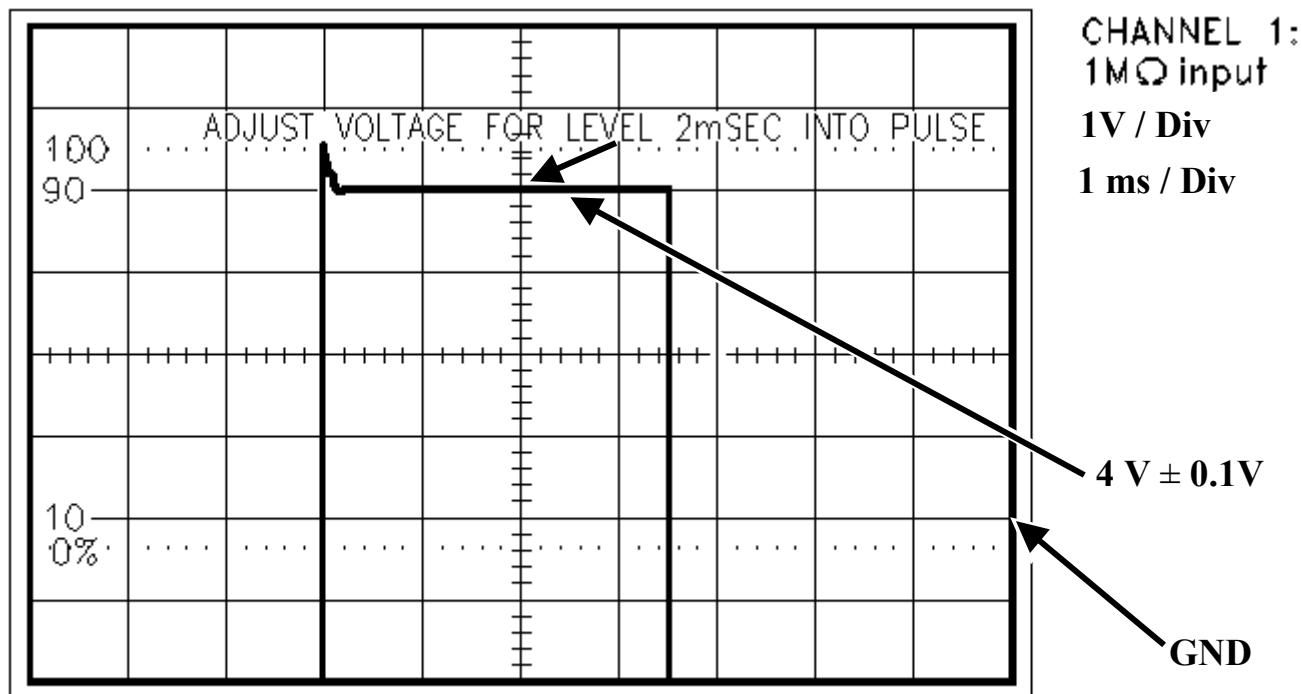
TABLE 3-3
TR DRIVER CIRCUIT TEST POINT DYNAMIC MEASUREMENTS

Function	Test point	Rating	Tolerance	Pot Adj.
TR Body (Loaded)	TP16	4.0 Vdc	± 0.1 Vdc	R13
TR Spectro (Loaded)	TP21	4.3 Vdc	± 0.1 Vdc	R18
TR Head (Loaded)	TP9	8.0 Vdc	± 0.1 Vdc	R26

Note

If the system has spectroscopy hardware then connect the Spectro TR switch box to the connectors on the head carriage before performing the next step. Otherwise, skip the next 2 steps.

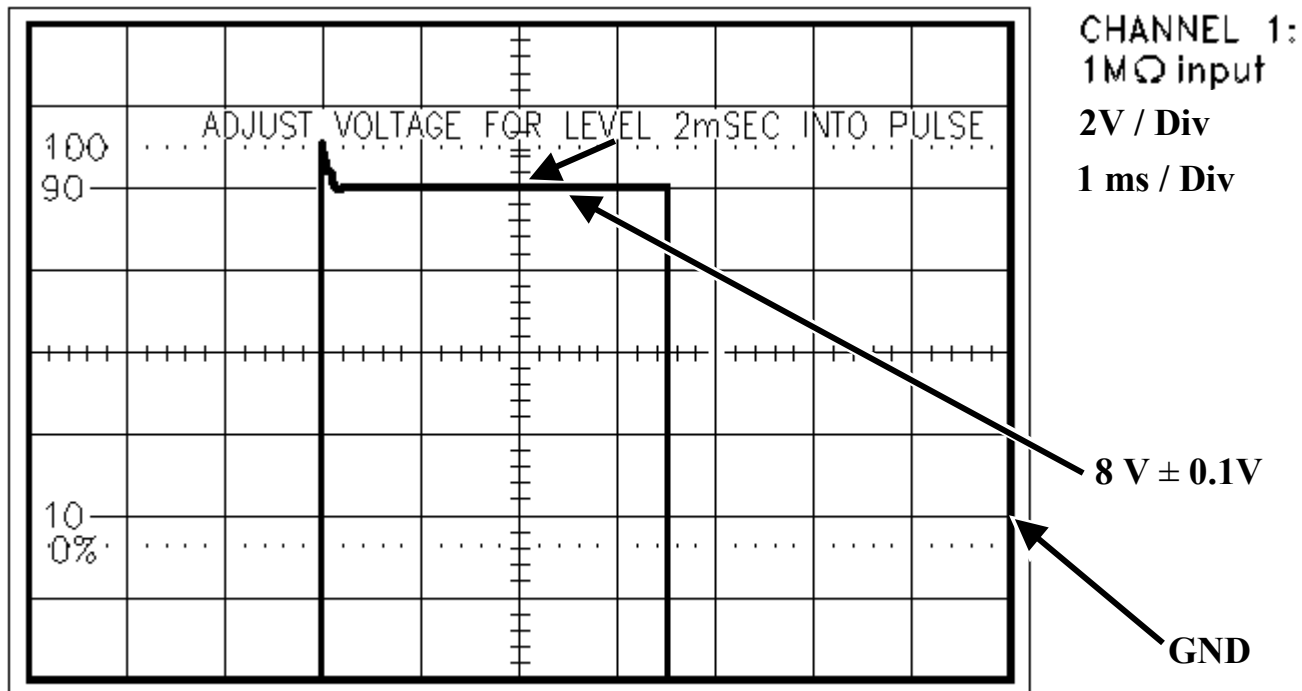
- e. Disconnect the scope lead from TP16 and attach it next to TP21 with the reference to the RFSC chassis. Trigger the signal so that there is one pulse exhibited on the scope.
- f. Refer to Table 3-3 and adjust R18 until the voltage 2 msec after the leading edge of the waveform at TP21 is $+4.3 \pm 0.1$ VDC with respect to the ground reference on the scope. The waveform will look exactly like that shown in Illustration 3-3 with the exception that the voltage is $+4.3 \pm 0.1$ VDC.



SCOPE BODY TR OR SPECTRO TR WAVEFORM
ILLUSTRATION 3-3

3-2 TR Driver Circuit Adjustments For Boards Without JP87 Pins 1 and 2 (Continued)

- g. Increase the channel 1 voltage input scale to 2 Volts / Division.
- h. Set the channel 1 input selector switch from DC to ground. Verify that the horizontal line trace is still centered exactly 2 divisions below the center horizontal x-axis line on the scope face. Reference Illustration 3-2.
- i. Set the channel 1 input selector switch from ground to DC.
- j. Disconnect the scope lead and move it to TP9 with the reference to the RFSC chassis. Trigger the signal so that there is one pulse exhibited on the scope.
- k. Refer to Table 3-3 and adjust R26 until the voltage 2 msec after the leading edge of the waveform at TP9 is $+8.0 \pm 0.1\text{VDC}$ with respect to the ground reference on the scope. See Illustration 3-4.



SCOPE HEAD TR WAVEFORM
ILLUSTRATION 3-4

- l. Disconnect the scope lead from TP9 and the reference and remove it from the RFSC chassis.

3-2 TR Driver Circuit Adjustments For Boards Without JP87 Pins 1 and 2 (Continued)

Note

It will be necessary to use an oscilloscope to measure the voltages at TP13, TP1, and TP18. A DVM or the same oscilloscope can be used to measure the steady-state, DC voltages at TP12, TP17, and TP19.

8. Reset the scope ground reference back to the center horizontal x-axis line and set channel 1 voltage input scale to 5 Volts / Division.
9. Verify that the loaded TR driver circuit operation on PIN Switch Driver Board as shown in Table 3-4.

TABLE 3-4
TR DRIVER LOADED CALIBRATION DATA

Step	Function	Location (Silk Screen)	Pot. Adj.	Specification
10	TR Driver (Body) (Loaded)	TP13 (W/TP59)		> Value @ TP12
11	TR Driver (Head) (Loaded)	TP1 (W/TP59)		> Value @ TP17
12	TR Driver (Spectro) (Loaded)	TP18 (W/TP59)		> Value @ TP19

10. Verify that the voltage at TP13, 2 msec after the leading edge of the waveform, is greater than the steady-state voltage at TP12, the body TR threshold. The voltage at TP13 is directly proportional to the body TR switch load current: **Current (I) of body TR = Voltage of TP13 ÷ 10**. If TP13 is not greater than TP12, recheck measurement at TP12.
11. Verify that the voltage at TP1, 2 msec after the leading edge of the waveform, is greater than the steady-state voltage measured at TP17, the head TR threshold. The voltage at TP1 is directly proportional to the head TR switch load current: **Current (I) of head TR = Voltage of TP1 ÷ 10**. If TP1 is not greater than TP17, recheck measurement TP17.

Note

If the system has spectroscopy hardware then connect the Spectro TR switch box to the connectors on the head carriage before performing the next step. Otherwise, skip the next step.

12. Verify that the voltage at TP18, 2 msec after the leading edge of the waveform, is greater than the steady-state voltage measured at TP19, the spectroscopy TR threshold. The voltage at TP18 is directly proportional to the Spectro TR switch load current: **Current (I) of spectro TR = Voltage of TP18 ÷ 10**. If Spectro TR Switch Box is not installed, then you will read less than 1.0 Vdc due to lack of current draw. If Spectro TR Switch is installed, and TP18 is not greater than TP19, recheck measurement at TP19.
13. Verify the dynamic operation of the TR drivers:
 - a. Select [**Pause**] at the keyboard to pause the scan.
 - b. Remove the scope and DMM leads from the RFSC chassis.
 - c. Move the RFSC top cover from the service to the closed position. Doing this will close the High Voltage Interlock Switch. The *HV ON* LED will illuminate.

3-2 TR Driver Circuit Adjustments For Boards Without JP87 Pins 1 and 2 (Continued)

- d. Move T/R-DD Faults Disable switch on the front of the RFSC to the OFF position. See Illustration 1-8.
- e. At the top of the keyboard select **[Start Scan]** to start the system pulsing.
- f. If TR errors are reported, and TR Switches are verified to be OK, recheck connections to J407, J408, and J409 at the rear of the RFSC module, and verify the sections of the calibration procedure relating to the TR driver circuits.
- g. When operation is satisfactory, select **[Stop Scan]**.
- h. Set the switch on the front of the ISE CERD (MR2A30A13) to "RF Out Normal" to enable excitation to the RF amplifier.
- i. Open the top cover of the RFSC and move it back into the service position. The *HV ON* LED will extinguish and the *T/R DD Faults* LED will illuminate.

4 – MULTICOIL DRIVER CIRCUIT ADJUSTMENTS

Systems with multicoil must be properly configured and adjusted. This is addressed in **Section 4-1 Multicoil Driver Circuit Adjustments For Systems With Multicoil**.

While there are no adjustments for systems without multicoil, it is important that the correct jumper settings are in place. This is addressed in **Section 4-2 Jumpers Settings For Systems Without Multicoil**. Reference Illustration 1-10 to locate the PIN Switch Driver Board test points. If needed, Table A-1 in Appendix A provides a summary of all the test points and corresponding adjustment potentiometers and test point specifications.

4-1 Multicoil Driver Circuit Adjustments For Systems With Multicoil

This section is divided between **Section 4-1-1 Multicoil Driver Output Voltage Adjustment**, which should be done at installation and whenever hardware is replaced, and **Section 4-1-2 Multicoil Functional Checks**, which should be done at installation and during troubleshooting.

4-1-1 Multicoil Driver Output Voltage Adjustment

1. Be sure that the T/R-DD Faults Disable switch is off (faults not disabled) on the PIN Switch Driver board. See Illustration 1-8.
2. Position jumpers JP57–60 to pins 1 and 2 (Transmit Simulation, Test Mode "C"). See Illustration 1-11. This sets a positive output voltage for the multicoil driver circuits which simulates the transmit mode.
3. Adjust the loaded multicoil driver output voltages on PIN Switch Driver Board as follows: Refer to Table 3-5.

TABLE 3-5
MULTI-COIL DRIVER CALIBRATION DATA

Step	Function	Location	Pot. adj.	Specification
3a	Multicoil 1 Driver (Loaded) Transmit	TP2 (W/TP59)	R133	+5.0 ± 0.1Vdc
3c	Multicoil 2 Driver (Loaded) Transmit	TP14 (W/TP59)		+5.0 ± 0.1Vdc
3c	Multicoil 3 Driver (Loaded) Transmit	TP20 (W/TP59)		+5.0 ± 0.1Vdc
3c	Multicoil 4 Driver (Loaded) Transmit	TP26 (W/TP59)		+5.0 ± 0.1Vdc
5a	Multicoil 1 Driver (Loaded) Receive	TP2 (W/TP59)	R129	-5.0 ± 0.1Vdc
5c	Multicoil 2 Driver (Loaded) Receive	TP14 (W/TP59)		-5.0 ± 0.1Vdc
5c	Multicoil 3 Driver (Loaded) Receive	TP20 (W/TP59)		-5.0 ± 0.1Vdc
5c	Multicoil 4 Driver (Loaded) Receive	TP26 (W/TP59)		-5.0 ± 0.1Vdc

- a. Measure the multicoil 1 driver output voltage at TP2 with respect to TGND (TP59).
 - b. Adjust R133 until the voltage at TP2 is +5.0 ± 0.1 Vdc.
 - c. Check multicoil 2, 3, 4 driver outputs at TP14, 20, 26, respectively, to ensure that they are at +5.0 ± 0.1 Vdc.
4. Move jumpers JP57–60 to pins 3 and 4 (Receive Simulation Mode "B"). See Illustration 1-11.

4-1-1 Multicoil Driver Output Voltage Adjustment (Continued)

5. Adjust the multicoil driver output voltage on PIN Switch Driver Board as shown in Table 3-5 above.
 - a. Measure the multicoil 1 driver output at TP2 with respect to TGND (TP59).
 - b. Adjust R129 until voltage at TP2 is -5.0 ± 0.1 Vdc.
 - c. Check multicoil 2, 3, 4 driver outputs at TP14, 20, 26 respectively to ensure that they are at -5.0 ± 0.1 Vdc.
6. Verify that jumpers JP57–60 on the PIN Switch Driver Board are across pins 5 and 6 (software control mode normal) for **four-channel multicoil** systems. For **two-channel multicoil** systems, verify that JP57 and JP59 are in software control mode (normal mode, pins 5 and 6 connected), and that JP58 and JP60 are in B mode (receive mode, pins 3 and 4 connected). See Illustration 1-11.

4-1-2 Multicoil Functional Checks

1. Be sure that the T/R-DD Faults Disable switch is off (faults not disabled) on the PIN Switch Driver board. See Illustration 1-8.
2. Position jumpers JP57–60 to pins 1 and 2 (Transmit Simulation, Test Mode "C"). See Illustration 1-11. This sets a positive output voltage for the multicoil driver circuits which simulates the transmit mode.
3. If the CTL coil is available then place it on the patient table and connect it to the phased array connector on the head carriage.
4. If the CTL coil is not available then verify that voltages at TP7, TP11, TP22, and TP28 are between -1.4 Vdc and -2.0 Vdc (unloaded value). Skip the Table 3-6 measurements.

Note

Two-channel Multicoil Setup - For a two-channel multicoil option, if a CTL coil is not available, use the Body Flex coil with the Dual Phased Array Adapter instead. With this configuration adjust only multicoil channel 1 and 3 while both are under load. Do not attempt to adjust multicoil channels 2 and 4 under load.

Note

For the following Table 3-6 measurements, do not assume the voltage measured is an absolute value. The polarities of the quantities are significant.

5. Verify that the loaded Multicoil Driver Error Detection circuit operation on PIN Switch Driver Board as shown in Table 3-6.

TABLE 3-6
MULTICOIL DRIVER/LOADED ERROR DETECTION DATA (TRANSMIT MODE)

Step	Function	Location	Pot. adj.	Specification
6	Multicoil 1 Driver (Loaded)	TP7 (W/TP59)		<TP8
7	Multicoil 2 Driver (Loaded)	TP11 (W/TP59)		<TP3
8	Multicoil 3 Driver (Loaded)	TP22 (W/TP59)		<TP25
9	Multicoil 4 Driver (Loaded)	TP28 (W/TP59)		<TP24

4-1-2 Multicoil Functional Checks (Continued)

6. When jumpers JP57–60 are configured for transmit mode (across pins 1 and 2), verify that the voltage at TP7 is less than the voltage at TP8. The voltage at TP7 is directly proportional to the multicoil 1 load current: **Current (I) of Multicoil 1 = Voltage of TP7 ÷ 5**. If TP7 is not less than the voltage at TP8, recheck measurements at TP8.
7. When jumpers JP57–60 are configured for transmit mode (across pins 1 and 2), verify that the voltage at TP11 is less than the voltage at TP3. The voltage at TP11 is directly proportional to the multicoil 2 load current: **Current (I) of Multicoil 2 = Voltage of TP11 ÷ 5**. If TP11 is not less than the voltage at TP3, recheck measurements at TP3.
8. When jumpers JP57–60 are configured for transmit mode (across pins 1 and 2), verify that the voltage at TP22 is less than the voltage at TP25. The voltage at TP22 is directly proportional to the multicoil 3 load current: **Current (I) of Multicoil 3 = Voltage of TP22 ÷ 5**. If TP22 is not less than the voltage at TP25, recheck measurements at TP25.
9. When jumpers JP57–60 are configured for transmit mode (across pins 1 and 2), verify the voltage at TP28 is less than the voltage at TP24. The voltage at TP28 is directly proportional to the multicoil 4 load current: **Current (I) of Multicoil 4 = Voltage of TP28 ÷ 5**. If TP28 is not less than the voltage at TP24, recheck measurements at TP24.
10. Position jumpers JP57–60 to pins 3 and 4 (Receive Simulation, Test Mode "B"). See Illustration 1-11. This sets a negative output voltage for the multicoil driver circuits which simulates the receive mode.
11. Verify that the loaded multicoil driver error detection circuit operation on PIN Switch Driver Board as shown in Table 3-7. (Keep the same coil connected as in the previous step.)

TABLE 3-7

MULTI-COIL DRIVER/LOADED ERROR DETECTION DATA (RECEIVE MODE)

Step	Function	Location	Pot. adj.	Specification
11a	Multicoil 1 Driver (Loaded)	TP7 (W/TP59)		<TP10 and >TP8
11b	Multicoil 2 Driver (Loaded)	TP11 (W/TP59)		<TP4 and >TP3
11c	Multicoil 3 Driver (Loaded)	TP22 (W/TP59)		<TP23 and >TP25
11d	Multicoil 4 Driver (Loaded)	TP28 (W/TP59)		<TP27 and >TP24

- a. When jumpers JP57–60 are configured for receive mode (across pins 3 and 4), verify that the voltage at TP7 is less than the voltage at TP10, but greater than the voltage at TP8. The voltage at TP7 is directly proportional to the multicoil 1 load current: **Current (I) of Multicoil 1 = Voltage of TP7 ÷ 5**. If TP7 is not between these voltages, recheck measurements at TP10 and TP8.
- b. When jumpers JP57–60 are configured for receive mode (across pins 3 and 4), verify the voltage at TP11 is less than the voltage at TP4, but greater than the voltage at TP3. The voltage at TP11 is directly proportional to the multicoil 2 load current: **Current (I) of Multicoil 2 = Voltage of TP11 ÷ 5**. If TP11 is not between these voltages, recheck measurements at TP3 and TP4.

4-1-2 Multicoil Functional Checks (Continued)

- c. When jumpers JP57–60 are configured for receive mode (across pins 3 and 4), verify that the voltage at TP22 is less than the voltage at TP23, but greater than the voltage at TP25. The voltage at TP22 is directly proportional to the multicoil 3 load current: **Current (I) of Multicoil 3 = Voltage of TP22 ÷ 5**. If TP22 is not between these voltages, recheck measurements at TP25 and TP23.
 - d. When jumpers JP57–60 are configured for receive mode (across pins 3 and 4), verify the voltage at TP28 is less than the voltage at TP27, but greater than the voltage at TP24. The voltage at TP28 is directly proportional to the multicoil 4 load current: **Current (I) of Multicoil 4 = Voltage of TP28 ÷ 5**. If TP28 is not between these voltages, recheck measurements at TP24 and TP27.
12. Reset any power monitor faults with the key switch.
 13. Verify that jumpers JP57–60 on the PIN Switch Driver Board are across pins 5 and 6 (software control mode normal) for **four-channel multicoil** systems. For **two-channel multicoil** systems, verify that JP57 and JP59 are in software control mode (normal mode, pins 5 and 6 connected), and that JP58 and JP60 are in B mode (receive mode, pins 3 and 4 connected). See Illustration 1-11.

4-2 Jumpers Settings For Systems Without Multicoil

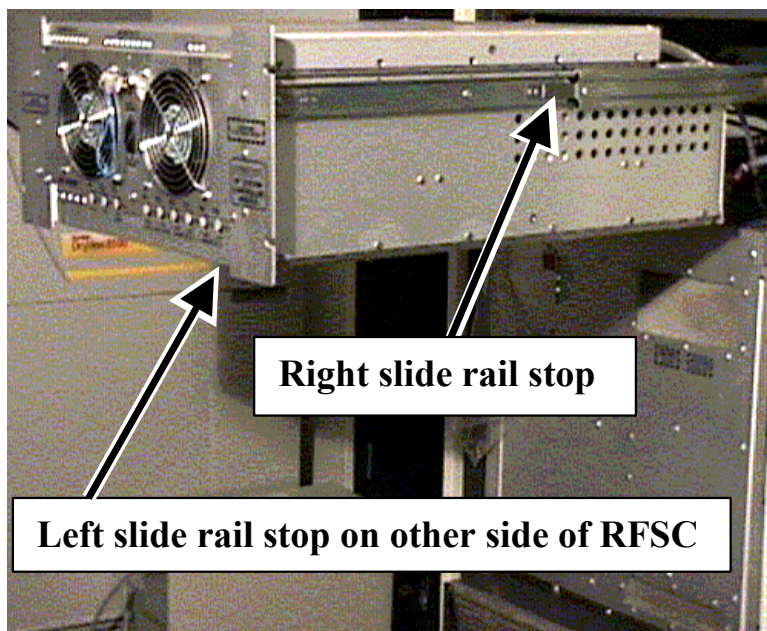
Verify that jumpers JP57-60 on the PIN Switch Driver Board are **across pins 3 and 4 Receive mode B for non-multicoil systems**. See Illustration 1-11.

5 – RESTORATION CHECK LIST

1. Replace the perforated rear shield and secure it to the top of the RFSC module with 3 screws. See Illustration 1-7.
2. At ISE CERD (MR2A30A13) enable excitation out to the RF amplifier by moving the RF toggle switch to the RF Out Normal position.
3. Verify that jumper JP87 is across pins 5 and 6 (software mode) and that jumpers JP5 and JP6 are across pins 1 and 2 (Normal mode "A" and Test Mode "B"). See Illustrations 1-13 and 1-12.
4. **Four-channel multicoil systems:** Verify that jumpers JP57–60 on the PIN Switch Driver Board are across pins 5 and 6 (software control mode normal). See Illustration 1-11.
5. **Two-channel multicoil systems:** Verify that JP57 and JP59 are in software control mode (normal mode, pins 5 and 6 connected), and that JP58 and JP60 are in B mode (receive mode, pins 3 and 4 connected). See Illustration 1-11.
6. **Non-multicoil systems:** Verify that jumpers JP57-60 on the PIN Switch Driver Board are in B mode (receive mode, pins 3 and 4 connected). See Illustration 1-11.
7. Close the top cover.
8. Install the 3 screws previously removed from above the fans on the front of the RFSC but do not tighten screws. See Illustration 1-4.
9. Install the 3 screws previously removed that secure the rear of the top cover to the perforated rear shield but do not tighten screws. See Illustration 1-6.
10. Secure the top cover to the RFSC chassis using the 4 captured screws. See Illustration 1-5.
11. Tighten the screws that were left loose in steps 9 and 10.

5 – RESTORATION CHECK LIST (CONTINUED)

12. Push the slide rail stops in on either side of the RFSC and, while minding the cables that connect to the rear of the RFSC, carefully slide the unit back into the cabinet. See Illustration 5-1.



DISENGAGING SLIDE RAIL STOPS

ILLUSTRATION 5-1

13. Secure the front of the RFSC to the cabinet with the 4 screws that had been previously removed. See Illustration 1-2.
14. Verify system functionality by successfully completing one head, body, and multicoil scan.
 - a. **Systems with 4-coil array multicoil:** Use, if possible, the CTL coil and perform the multicoil scan in the sagittal plane.
 - b. **Systems with 2-coil array multicoil:** Use, if possible, a 2-coil array when performing the multicoil scan.

APPENDIX – A

TABLE A-1
SUMMARY OF PIN SWITCH DRIVER TEST POINTS

Function	Location	Pot. Adj.	Meas. Value	Specification
–15 V _A Power Supply	TP56 (W/ TP58)			
+15 V _A Power Supply	TP51 (W/ TP58)			
+5 Vdc Power Supply	TP54 (W/ TP58)			+5.0 ± 0.25 Vdc
–15 Vdc Power Supply	TP57 (W/ TP58)			–15.0 ± 0.2 Vdc
+15 Vdc Power Supply	TP55 (W/ TP58)			+15.0 ± 0.2 Vdc
+500/1000 Vdc Power Supply	TP62 (W/ TP58)			+500 ± 25 Vdc +1000 ± 50 Vdc
Direct Drive Output (J402 Connected)	TP40 (W/TP56)			1.0 ± 0.2 Vdc
Dyn. Dis. Driver #1 (J403 Connected)	TP32 (W/TP56)			+1.0 ± 0.2 Vdc
Dyn. Dis. Driver #2 (J401 Connected)	TP46 (W/TP56)			+1.0 ± 0.2 Vdc
Direct Drive Output Voltage	TP38 (W/TP59)			–2.5 ± 1.0 Vdc
Dyn. Dis. Driver #1 (Loaded)	TP31 (W/TP59)			–4.0 ± 1.5 Vdc
Dyn. Dis. Driver #2 (Loaded)	TP46 (W/TP59)			–4.0 ± 1.5 Vdc
Direct Drive Open Circuit Detect	TP44 (W/TP59)	R207		Value @ (TP45 – 2.0) ± 0.1 Vdc
Dyn. Dis. Driver #1 Open Cir. Detect	TP34 (W/ TP59)	R147		Value @ (TP33 – 2.0) ± 0.1 Vdc
Dyn. Dis. Driver #2 Open Cir. Detect	TP48 (W/ TP59)	R228		Value @ (TP47 – 2.0) ± 0.1 Vdc
Direct Drive Output Voltage	TP38 (W/TP59)			Value @ TP62+0/– 10 Vdc
Dyn. Dis. Driver #2 Output Voltage	TP46 (W/TP59)			Value @ TP62 +0/– 10.0 Vdc
TR Driver (Body) (Loaded)	TP16 (W/TP59)	R13		+4.0 ± 0.1 Vdc
TR Driver (Head) (Loaded)	TP9 (W/TP59)	R26		+8.0 ± 0.1 Vdc
TR Driver (Spectro) (Loaded)	TP21 (W/TP59)	R18		+4.3 ± 0.1 Vdc
TR Driver (Body) (Loaded)	TP13 (W/TP59)			> Value @ TP12
TR Driver (Head) (Loaded)	TP1 (W/TP59)			> Value @ TP17
TR Driver (Spectro) (Loaded)	TP18 (W/TP59)			> Value @ TP19
Head TR Current Output	TP1 (W/TP59)			0.0 ± 0.1 Vdc
Head TR Voltage Output	TP9 (W/TP59)			–13.75 ± 1.5 Vdc
Body TR Current Output	TP13 (W/TP59)			0.0 ± 0.25 Vdc
Body TR Voltage Output	TP16 (W/TP59)			–13.75 ± 1.5 Vdc
Spectro TR Current Output	TP18 (W/TP59)			0.0 ± 0.25 Vdc
Spectro TR Voltage Output	TP21 (W/TP59)			–13.75 ± 1.5 Vdc
TR Driver Circuit	TP16	R13		+4.0 ± 0.5 Vdc
TR Driver Circuit	TP9	R26		+8.0 ± 0.5 Vdc
TR Driver Circuit	TP21	R18		+4.0 ± 0.5 Vdc
Multicoil 1 (Loaded) (Transmit)	TP2 (W/TP59)	R133		+5.0 ± 0.1 Vdc
Multicoil 2 (Loaded) (Transmit)	TP14 (W/TP59)			+5.0 ± 0.1 Vdc
Multicoil 3 (Loaded) (Transmit)	TP20 (W/TP59)			+5.0 ± 0.1 Vdc
Multicoil 4 (Loaded) (Transmit)	TP26 (W/TP59)			+5.0 ± 0.1 Vdc
Multicoil 1 (Loaded) (Receive)	TP2 (W/TP59)	R129		–5.0 ± 0.1 Vdc
Multicoil 2 (Loaded) (Receive)	TP14 (W/TP59)			–5.0 ± 0.1 Vdc
Multicoil 3 (Loaded) (Receive)	TP20 (W/TP59)			–5.0 ± 0.1 Vdc

Multicoil 4 (Loaded) (Receive)	TP26 (W/TP59)			-5.0 ± 0.1 Vdc
Multicoil 1 Error Detect	TP7 (W/TP59)			<TP8 (Loaded) (Transmit)
Multicoil 2 Error Detect	TP11 (W/TP59)			<TP3 (Loaded) (Transmit)
Multicoil 3 Error Detect	TP22 (W/TP59)			<TP25 (Loaded) (Transmit)
Multicoil 4 Error Detect	TP28 (W/TP59)			<TP24 (Loaded) (Transmit)
Multicoil 1 Error Detect	TP7 (W/TP59)			<TP10 and >TP8 (Loaded) (Receive)
Multicoil 2 Error Detect	TP11 (W/TP59)			<TP4 and >TP3 (Loaded) (Receive)
Multicoil 3 Error Detect	TP22 (W/TP59)			<TP23 and >TP25 (Loaded) (Receive)
Multicoil 4 Error Detect	TP28 (W/TP59)			<TP27 and >TP24 (Loaded) (Receive)

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 24, 1998	J. Saperstein	Initial conversion from Toolbook to Word.
1	Nov 9, 2000	D. Thome	Added section 3-2 for boards without JP87. Corrected formatting issues. Added Quick Reference for boards without JP87. Added section 4-2 for non-multicoil systems.
2	February 28, 2001	D. Thome	Re-wrote for JP87 missing. Restructured and streamlined calibration process. Removed Quick Check Section.
3	March 8, 2001	D. Thome	Correct errors in Tables 3-4, 3-5. Added comments to check pins 5 and 6 on JP87 jumpers without pins 1 and 2. Corrected annotation on Illustration 1-13.
4	March 22, 2001	D. Thome	Updated Spectro bias voltage to 4.3 +/- 0.1 VDC.
5	Nov 13, 2003	D. Thome	Removed incorrect references.